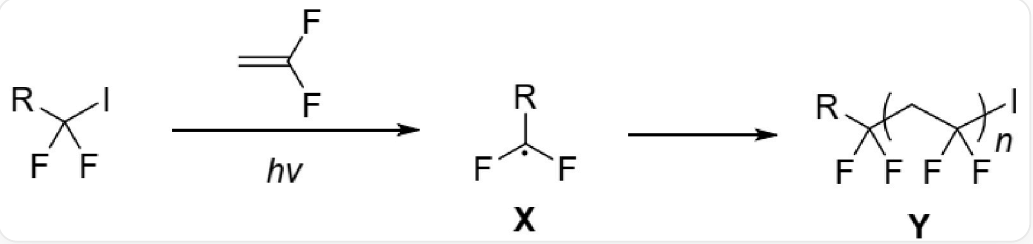
Question

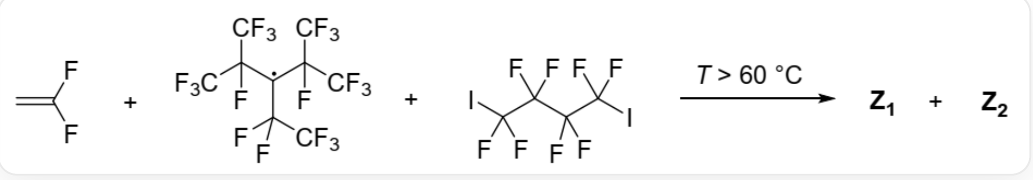
Alkyl iodides (RCF_2I) can undergo polymerization with 1,1-difluoroethylene, as shown in the figure below:

In this process, alkyl iodides act as free radical initiating species, undergoing homolytic cleavage of the C-I bond under light conditions to generate the carbon radical $\mathbf{X}$. $\mathbf{X}$ initiates the polymerization of 1,1-difluoroethylene, resulting in the polymeric compound $\mathbf{Y}$.



FC(F)(I)[R] undergoes polymerization with C=C(F)F under light conditions, first producing the radical $\mathbf{X}$: F[C](F)[R], and then obtaining the polymer $\mathbf{Y}$: The basic structure is `FC(F)(CC(F)(F)I)[R]` (when there is only one repeating unit), where `[*]CC(F)(F)[*]` is the repeating unit

PPFR (structure `FC(C(F)(F)F)(C(F)(F)F)[C](C(F)(C(F)(F)F)F)C(F)(C(F)(F)F)C(F)(F)F`, with the central carbon atom being a radical) is a reagent that can generate trifluoromethyl radicals upon heating. The free radical polymerization shown in the figure below yields two polymers, $\mathbf{Z1}$ and $\mathbf{Z2}$, in the presence of a catalytic amount of PPFR. It is known that $\mathbf{Z1}$ and $\mathbf{Z2}$ have different end groups and the number average molecular weight of $\mathbf{Z1}$ is approximately twice that of $\mathbf{Z2}$.



1,1-Difluoroethylene and PPFR and `IC(F)(F)C(F)(F)C(F)(F)C(F)(F)I` undergo polymerization at temperatures above 60 degrees Celsius to produce two polymers $\mathbf{Z1}$ and $\mathbf{Z2}$

The following statements are made:

1. In the chain transfer step of the polymer $\mathbf{Y}$ reaction, the polymer chain can abstract an iodine atom from the alkyl iodide to generate a new $\mathbf{X}$.

2. $\mathbf{Z1}$ has the same end groups on both ends.

3. $\mathbf{Z1}$ may have a center of symmetry.

4. $\mathbf{Z2}$ does not contain iodine.

Select the sum of the coefficients of all correct statements:

A. No correct statement

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. 7

I. 8

J. 9

K. 10

Answer

Correct Answer: G

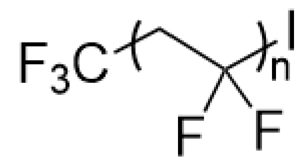
Detailed Explanation

Under illumination, the C-I bond of alkyl iodides undergoes homolytic cleavage, generating initiating radicals $RCF_2\cdot$ (i.e., $\mathbf{X}$) and an iodine radical. $\mathbf{X}$ attacks 1,1-difluoroethylene to initiate polymerization. Subsequently, chain transfer can occur, where the polymer radical attacks RCF_2I to abstract an iodine atom, generating new $\mathbf{X}$; or the polymer radical reacts with a molecule of iodine radical, terminating the reaction.

CHECKPOINT 1 PTS

The polymer radical can abstract the iodine atom of alkyl iodide to generate new $\mathbf{X}$, statement 1 is correct

In the reactions producing $\mathbf{Z1}$ and $\mathbf{Z2}$, because the free radical polymerization has a narrow molecular weight distribution, and the number-average molecular weight of $\mathbf{Z1}$ is approximately twice that of $\mathbf{Z2}$, then $\mathbf{Z1}$ likely has two polymer segments. First, PPFR is heated to generate trifluoromethyl radicals, which then attack 1,1-difluoroethylene, generating a new radical `F[C](F)CC(F)(F)F`, initiating polymerization until the polymer radical abstracts an iodine atom from `IC(F)(F)C(F)(F)C(F)(F)C(F)(F)I`, generating a new radical `IC(F)(C(F)(C(F)([C](F)F)F)F)F`. At this point, the polymerization of the original polymer terminates, resulting in a basic polymer structure of `FC(F)(I)CC(F)(F)F` (when there is only one repeating unit), where the repeating unit is: `FC(F)([*])C[*]`, which should be $\mathbf{Z2}$, containing an iodine atom.

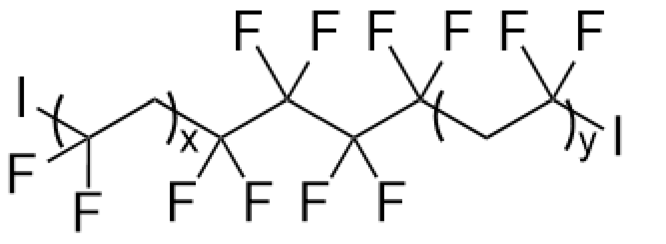


The figure shows the structure of $\mathbf{Z2}$, the basic structure is `FC(F)(I)CC(F)(F)F` (when there is only one repeating unit), the repeating unit is: `FC(F)([*])C[*]`

CHECKPOINT 1 PTS

$\mathbf{Z2}$ needs to abstract an iodine atom during chain termination, therefore it contains iodine, statement 4 is incorrect

The newly generated radical then reacts with 1,1-difluoroethylene to generate the radical `IC(F)(C(F)(C(F)(C(F)(C[C](F)F)F)F)F)F`, initiating polymerization. Polymerization terminates when another iodine atom is abstracted from `IC(F)(F)C(F)(F)C(F)(F)C(F)(F)I`. Since `IC(F)(F)C(F)(F)C(F)(F)C(F)(F)I` has two iodine atoms, it can be abstracted twice to initiate two segments of polymerization. Then the final basic structure of the polymer is `FC(F)(C(F)(C(F)(C(F)(CC(F)(I)F)F)F)F)CC(F)(F)I` (when there is only one repeating unit), with two repeating units `FC(F)([])C[]`. Because it has two polymerization segments, the number-average molecular weight is approximately twice that of $\mathbf{Z2}$, which is $\mathbf{Z1}$, and the structure is shown in the figure below:



The basic structure of $\mathbf{Z1}$ is `FC(F)(C(F)(C(F)(C(F)(CC(F)(I)F)F)F)F)CC(F)(F)I` (when there is only one repeating unit), with two repeating units `FC(F)([*])C[*]`

Since the end groups of **Z1** are all iodine atoms, the end groups are the same.

CHECKPOINT 1 PTS

The end groups of **Z1** are all iodine atoms, the end groups are the same, statement 2 is correct

The center of **Z1** is `FC(C(F)(C(F)(C(F)([])F)F)F)([])F` (the asterisk part connects to other parts of the polymer), which has a center of symmetry. If the degrees of polymerization of the two polymer segments are the same, then **Z1** can have a center of symmetry.

CHECKPOINT 1 PTS

If the degrees of polymerization of the two polymer segments are the same, then **Z1** can have a center of symmetry, statement 3 is correct

The correct statements are 1, 2, and 3, the sum is 6, choose G.

In this question, why is the species that initiates polymerization `F[C](F)CC([R])(F)F` (the radical is the carbon atom of the original difluoroethylene connected to two fluorine atoms, R represents other groups) instead of `FC(F)(C(F)(F)[R])[C]([H])[H]` where the radical is the carbon atom of the original difluoroethylene connected to two hydrogen atoms, R represents other groups: These two radicals do not have particularly large differences in stability, but when a carbon atom is connected to two fluorine atoms, the C-I bond is weakened. Therefore, if it is the `FC(F)(C(F)(F)[R])[C]([H])[H]` radical, it can abstract an iodine atom to generate a relatively stable C-I bond instead of further generating new radicals, which will terminate the polymerization. Only the C-I bond generated by the `F[C](F)CC([R])(F)F` radical is relatively weak, which can allow the polymerization to continue by further generating new radicals.

CHECKPOINT 1 PTS

In this question, the species that initiates polymerization is the `F[C](F)CC([R])(F)F` radical